

1 Probability and Statistics

- (RANDOM VARIABLE) Suppose you flip a fair coin that can land either heads (H) or tails (T). If it lands heads, the value 1 is assigned as the outcome; if it lands tails, the value -1 is assigned. Let X be the random variable representing the outcome.
 - What is the sample space of the experiment?
 - Taking the event space to be the power set of the sample space, what are all the possible events?
 - What is the probability of each event?
 - What is a random variable? Define X as a function on the sample space.
- (JOINT, MARGINAL, AND CONDITIONAL DISTRIBUTIONS) The joint probability mass function of (X, Y) is given by

	$Y = 0$	$Y = 1$
$X = 0$	0.10	0.20
$X = 1$	0.30	0.40

- Verify that the table defines a valid joint distribution.
 - Find the marginal distributions of X and Y .
 - Find $\Pr(X = 1 | Y = 1)$ and $\Pr(Y = 1 | X = 0)$.
 - Compute $\mathbb{E}(X)$, $\text{Var}(X)$, $\mathbb{E}(Y)$, and $\text{Var}(Y)$.
 - Compute $\text{Cov}(X, Y)$. Are X and Y independent?
- (CONTINUOUS RANDOM VARIABLES) Suppose that the random variable U has probability density function

$$f_U(u) = \begin{cases} 2u, & 0 \leq u \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

- Verify that f_U is a valid probability density function (PDF).
- Derive the cumulative distribution function (CDF) $F_U(u)$, including its values outside $[0, 1]$.
- Compute $\Pr(1/4 < U \leq 3/4)$ using both f_U and F_U .
- What is $\Pr(U = 1/2)$?
- Find the median m satisfying $F_U(m) = 1/2$.
- Compute $\mathbb{E}(U)$, $\mathbb{E}(U^2)$, and $\text{Var}(U)$.

Now suppose that the random variable V has CDF

$$F_V(v) = \begin{cases} 1 - e^{-v}, & v \geq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

- (g) Derive the PDF $f_V(v)$.
- (h) Compute $\Pr(V > 2)$.
- (i) Find the 90th percentile of V .
- (j) Compute $\mathbb{E}(V)$, $\mathbb{E}(V^2)$, and $\text{Var}(V)$.

2 Linear Algebra and Matrix Algebra

Let

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}. \quad (3)$$

- 1. What are the dimensions of A and A' ? A' denotes the transpose of A .
- 2. What is the rank of A ?
- 3. Compute $A'A$.
- 4. Is $A'A$ invertible? If so, compute $(A'A)^{-1}$.
- 5. What property of A guarantees that $A'A$ is invertible?

3 Calculus: Taylor Expansions

Let $g(x) = \log(x)$ for $x > 0$.

- 1. Compute $g'(x)$, $g''(x)$, and $g'''(x)$.
- 2. State precisely what $o(h^k)$ and $O(h^k)$ mean as $h \rightarrow 0$ for any fixed positive integer k . Also explain intuitively what these notations mean.
- 3. Write the first-order and second-order Taylor expansions of $g(x)$ around $x = 1$. Precisely, define $h = x - 1$ and write the Taylor expansions of $g(1+h)$. For each expansion, express the remainder using both little- o notation and the sharp big- O order.
- 4. Use both expansions to approximate $\log(1.1)$.
- 5. Which approximation is more accurate? Relate the approximation errors to the remainder orders.

4 Calculus: Optimization

Let $q : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$q(x, y) = 4x + 2y - x^2 - xy - y^2. \quad (4)$$

1. (OPTIMALITY CONDITIONS)
 - (a) State the *first-order necessary condition* for an interior local maximum of a differentiable function.
 - (b) State the *second-order sufficient condition* for an interior point to be a local maximum of a twice-differentiable function.
2. Compute the gradient of q and solve the first-order conditions.
3. Compute the Hessian of q and determine whether it is positive definite, negative definite, or indefinite.
4. Use the second-order condition to classify the stationary point. Is it also the unique global maximizer?